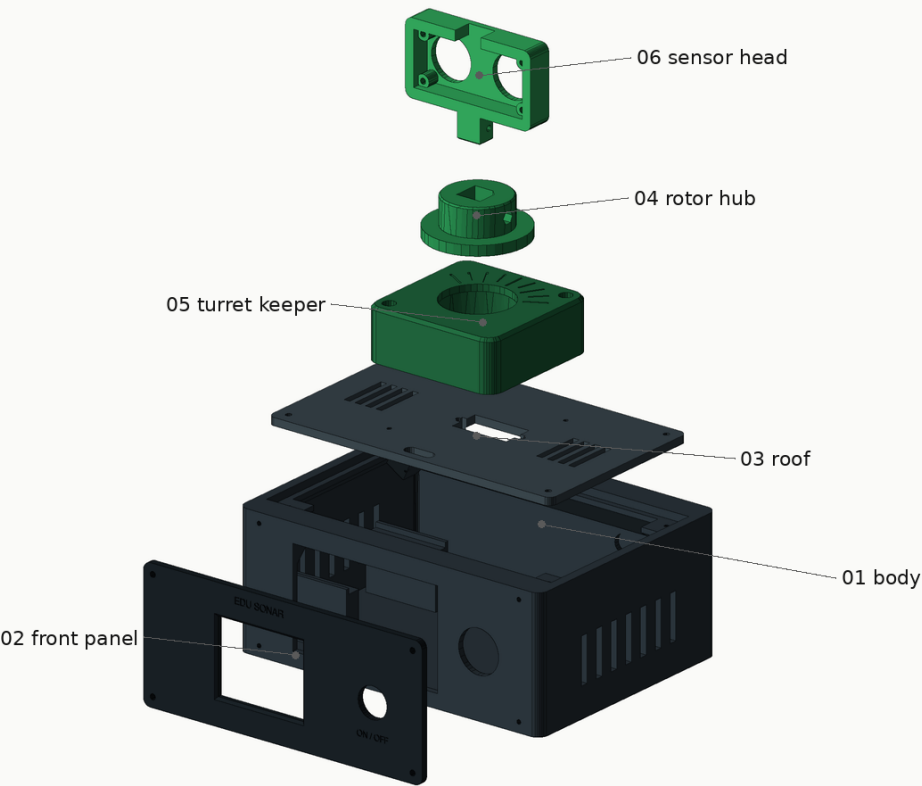


Ultrasonic Radar Scanner

Build packet — Radar V5.1 hardware baseline · repository release 5.2.0 · generated 2026-10-02

Radar V5.1 - printed parts (exploded)

Rendered from cad/build_radar_v5.py - geometry only, not a photo of a printed unit



Honest scope. An educational ultrasonic (sonar) instrument with a radar-style display — not radio-frequency radar and not a safety, obstacle-avoidance or people-detection device. It sweeps about 120°; the angle shown is the commanded servo position. At the time this packet was generated, no unit had been printed, wired or powered — your build is part of the validation.

Unit / serial	
Builder	
Start date	
Printer + filament	
Firmware (version / SHA-256)	
Notes	

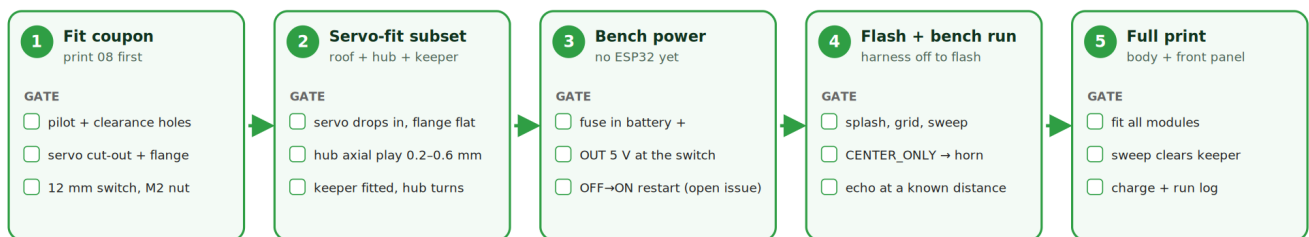
1 · Safety rules (read before buying anything)

- ☐ The **3 A fuse on the battery positive lead is mandatory**; fit it before anything else touches the cell.
- ☐ Never connect a bare cell to the ESP32 VIN or to the servo — only the charger module's 5 V output feeds the scanner.
- ☐ Never solder to the cell; never put two cells in series; use a **protected** 18650 (≥ 2.1 A charge, ≥ 3 A discharge).
- ☐ Secure the **holder**, not the cell; nothing sharp against the cell wrapping; heat-shrink every power joint.
- ☐ HC-SR04 ECHO (5 V) reaches GPIO26 **only** through the 2.2 k Ω / 3.3 k Ω divider.
- ☐ Unplug the whole ESP32 harness before connecting a programming cable; never USB and external 5 V together.
- ☐ Do not charge unattended while prototyping; stop if anything gets hot, smells, swells or hisses.

Build stages — each has a gate; record results before moving on

Build stages — do not skip a gate

Each stage has a pass/fail gate. Record the result in docs/VALIDATION.md before moving right. A failed gate means: adjust the named VERIFY variable, re-render, repeat.



2 · Parts checklist

What you need — the parts kit

Generated from docs/BOM.md by scripts/export_mfg.py · quantities in brackets · full details and checks in manufacturing/bom.csv



Electronics

- [1] ESP32-WROOM-32 DevKit, 30-pin
- [1] HC-SR04 ultrasonic sensor
- [1] SG90 positional servo + horn
- [1] 1.8" ST7735S LCD (Waveshare)



Power and protection

- [1] DFRobot DFR1026 charger/boost
- [1] USB-C breakout, 5.1 kΩ CC×2
- [1] Protected 18650 cell
- [1] 18650 holder with leads
- [1] 12 mm latching switch ≥2 A
- [1] 3 A fuse + inline holder
- [1] 1000 μF ≥10 V capacitor



Passive parts

- [1] 2.2 kΩ 1 % resistor
- [1] 3.3 kΩ 1 % resistor



Wiring and mounting

- [~20] jumper leads + 22 AWG wire
- [≥1] Keyed 2-pin plug (J1)
- [1 lot] Heat-shrink, tape, ties, feet



Cables

- [1] 5 V 3 A USB-C supply + cable
- [1] USB data cable (ESP32)



Consumables

- [~260 g max] PLA filament



Fasteners (M2 only)

- [2] M2×6 screw
- [12] M2×8 screw
- [4] M2×12 screw
- [1] M2×20 screw
- [6] M2 hex nut
- [8] M2 washer
- [1] M2 hand tap



Optional (live link)

- [1] 3.3 V USB-serial adapter

Plus the six printed parts and the fit coupon (manufacturing/printed_parts.csv). Recheck every listing before buying; safety parts are not optional.

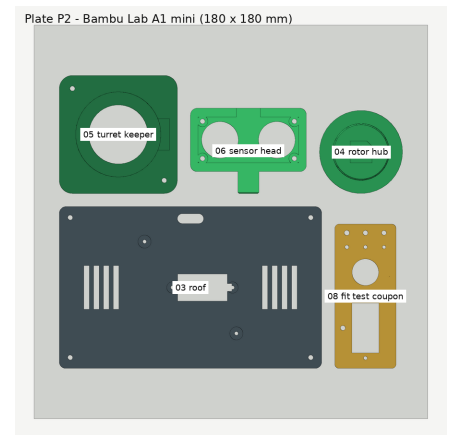
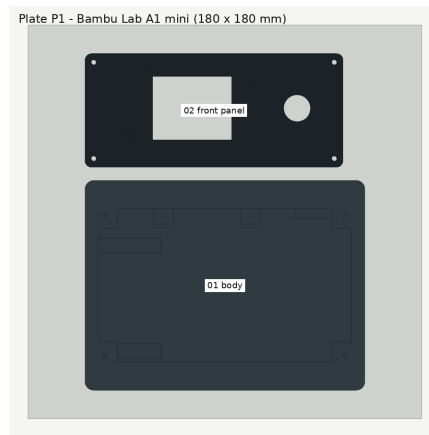
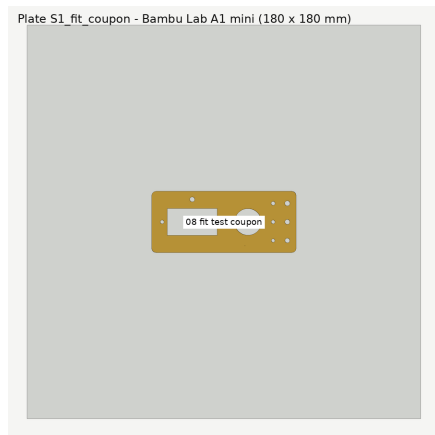
✓	ID	Qty	Part	Check before you buy
☐	E1	1	ESP32-WROOM-32 development board ("DevKit"), classic 30-pin, headers already soldered	Module marking says WROOM-32 (not WROVER); pins labelled GPIO16/17/18/23/26/27/33/14; a USB data port. Do not substitute an ESP32-C3, ESP32-S3 or a WROVER board without re-checking every pin and the firmware — GPIO16/17 are used for PSRAM on WROVER.
☐	E2	1	HC-SR04 ultrasonic sensor	5 V part with 4-pin header VCC/TRIG/ECHO/GND; four corner mounting holes (the sensor head expects them — SR04_HOLE_X/Y in the CAD are VERIFY values).
☐	E3	1	SG90-size positional micro servo with its factory double-arm horn	180°-type positional servo, not continuous-rotation ("360°") — those cannot hold an angle. Keep the factory horn and spline.
☐	E4	1	Waveshare 1.8" SPI LCD, ST7735S, 128×160	Pins VCC/GND/DIN/CLK/CS/DC/RST/BL; board 56.5 × 34 mm (check your revision).
☐	E5	1	DFRobot DFR1026 charge/discharge module (5 V / 2 A output)	Accepts 4.6–5.4 V input, ~2 A charge. It has no USB connector — needs E6. See the open issue before committing to it.
☐	E6	1	USB-C female power-input breakout with 5.1 kΩ pull-downs on both CC1 and CC2	The two 5.1 kΩ resistors (one per CC pin) are what make a USB-C charger switch on 5 V. No Power Delivery (PD) trigger board — this project must never receive 9 V or 12 V.
☐	E7	1	Protected 18650 lithium-ion cell, ~2600–3500 mAh	Protected (built-in protection circuit). Check the datasheet for ≥2.1 A charge and ≥3 A continuous discharge — capacity alone is not enough. Buy from a reputable supplier; protected cells are slightly longer than bare ones, so check the holder fits.
☐	E8	1	Wired single-cell 18650 holder	Flying leads (no soldering to the cell), fits a protected cell's length.
☐	E9	1	12 mm latching push-on/push-off button, 2 terminals, single-pole single-throw (SPST)	Rated ≥2 A at 5 V DC (check the DC rating, not only the AC one). Panel hole 12 mm; fascia is 2.6 mm thick.
☐	E10	1	3 A fuse + insulated inline holder	On battery positive, as close to the holder as practical. Blade (ATM/mini) or 5 × 20 mm glass both fine.
☐	E11	1	1000 μF electrolytic capacitor, ≥10 V (16 V gives margin)	Polarised: stripe = negative. Mounted at the servo end of its supply branch.
☐	E12	1	2.2 kΩ resistor, 1 %, ¼ W	ECHO divider R1.
☐	E13	1	3.3 kΩ resistor, 1 %, ¼ W	ECHO divider R2.

✓	ID	Qty	Part	Check before you buy
☐	E14	~20	Jumper leads, mostly female-female, plus short 22-24 AWG stranded power wire	Power branches (charger → switch → load) in 22 AWG; signal leads can be jumpers.
☐	E15	≥1	Keyed two-pin power disconnect (J1)	Polarised so it cannot be reversed (for example JST-XH 2.54 mm). Lets the ESP32's power lead unplug for programming.
☐	E16	1 lot	Heat-shrink, double-sided foam tape, 2 battery straps or zip ties (≤3.6 mm wide), 4 rubber feet	Straps go round the holder, not the bare cell.
☐	E17	1	5 V / 3 A USB-C wall supply + USB-C cable	Plain 5 V is enough; a PD charger also works because the breakout only asks for 5 V.
☐	E18	1	USB data cable for the ESP32 board	Micro-USB or USB-C to match E1. Charge-only cables will not program.
☐	E19	~260 g max	PLA filament	257 g is the solid-volume upper bound for all required parts plus the coupon (from cad/geometry_report.json); your slicer's figure with walls + infill will be lower.
☐	E20	1 (optional)	3.3 V-logic USB-to-serial adapter + 2 female jumper leads	Only for the live console. CP2102, CH340 or FT232 type, voltage set to 3.3 V. Only its RX and GND are used; VCC stays unconnected. See LIVE_LINK.md.
☐	F1	2	M2×6 screw	horn → rotor hub (from below, through the horn arm holes)
☐	F2	12	M2×8 screw	4 roof → body, 4 front panel → body, 2 servo flange → roof, 2 turret keeper → roof
☐	F3	4	M2×12 screw	HC-SR04 board → sensor head (nuts on the back)
☐	F4	1	M2×20 screw	cross-bolt through the rotor hub and the sensor-head mast
☐	F5	6	M2 hex nut	4 sensor + 1 cross-bolt + 1 spare — buy extras
☐	F6	8	M2 washer	6–10 under screw heads on printed plastic
☐	F7	1	M2 hand tap	cuts threads in the 1.70 mm printed pilot holes

3 · Printed parts

Files: **cad/3mf/** (Bambu A1 mini plates), **cad/stl/** (already in print orientation), **cad/step/** (editable). Print the fit coupon first and adjust any VERIFY value in cad/build_radar_v5.py before the long prints. PLA, 0.20 mm layers, 3 walls, 15–20 % gyroid, **no supports**.

✓	Part	Role	Plate	Print size (mm)	PLA ≤ g	Orientation
☐	01_body	required	P1	128.0 x 96.0 x 52.0	141.9	floor on the plate; 45-degree deck needs no supports
☐	02_front_panel	required	P1	118.0 x 52.0 x 6.6	16.4	viewer face on the plate; locators up
☐	03_roof	required	P2	120.0 x 74.0 x 8.0	30.6	top face on the plate; bosses up
☐	04_rotor_hub	required	P2	38.0 x 38.0 x 15.0	10.7	flange on the plate; horn nest bridges
☐	05_turret_keeper	required	P2	54.0 x 54.0 x 20.0	42.5	lip/top face on the plate
☐	06_sensor_head	required	P2	53.0 x 39.0 x 10.0	9.5	face-down; mast lies on the plate
☐	07_keeper_shim_0p4mm	optional	P3	54.0 x 54.0 x 0.4	0.8	flat; two layers at 0.2 mm
☐	08_fit_test_coupon	print first - test piece	P2	66.0 x 28.0 x 3.0	5.2	flat; print this first



Fit coupon (stage 1)

✓	Coupon test	Pass when	Size chosen
☐	Pilot holes 1.60 / 1.70 / 1.80 mm	smallest that taps M2 cleanly and holds	
☐	Clearance holes 2.15 / 2.25 / 2.35 mm	smallest an M2 drops through	
☐	Servo cut-out + flange pilots	servo drops in, flange flat, holes align	
☐	12 mm switch hole	switch fits, nut tightens	
☐	M2 nut pocket	nut presses in and stays	

4 · Wiring — point to point

Cut, label and fit each wire, then tick it. Follow the **labels** on the modules, not wire colours. Power wires first (stage 3, no ESP32), then signals (stage 4). Source of truth: docs/WIRING.md.

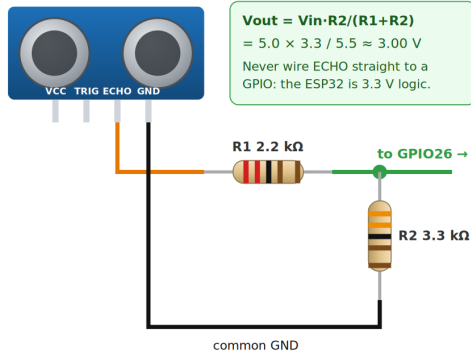
✓	Wire	From	To	Net	Wire type	Notes
☐	W01	USBC.VBUS	DFR1026.VIN	USB_5V	22 AWG	charger input (4.6–5.4 V)
☐	W02	USBC.GND	DFR1026.GND_IN	USB_GND	22 AWG	charger input ground pad
☐	W03	CELL.+	FUSE.A	BAT_RAW	22 AWG	3 A fuse, insulated holder, mandatory
☐	W04	FUSE.B	DFR1026.BAT+	BAT_POS	22 AWG	
☐	W05	CELL.-	DFR1026.GND_BAT	BAT_NEG	22 AWG	battery ground pad only; do not tie elsewhere
☐	W06	DFR1026.OUT_5V	SW.1	OUT_5V	22 AWG	latching switch, rated ≥ 2 A at 5 V
☐	W07	SW.2	ESP32.VIN	LOAD_5V	22–24 AWG	via keyed disconnect J1 pin 1; own branch from the star point
☐	W08	SW.2	SERVO.RED	LOAD_5V	22 AWG	own short branch, separate from the ESP32 branch
☐	W09	SW.2	C1.+	LOAD_5V	22 AWG to component lead	1000 μ F ≥ 10 V, mounted at the servo end; mind polarity
☐	W10	SW.2	SR04.VCC	LOAD_5V	jumper	
☐	W11	DFR1026.GND_OUT	ESP32.GND	GND	22–24 AWG	via keyed disconnect J1 pin 2
☐	W12	DFR1026.GND_OUT	SERVO.BROWN	GND	22 AWG	brown or black lead
☐	W13	DFR1026.GND_OUT	C1.-	GND	22 AWG to component lead	stripe side
☐	W14	DFR1026.GND_OUT	SR04.GND	GND	jumper	
☐	W15	DFR1026.GND_OUT	LCD.GND	GND	jumper	
☐	W16	DFR1026.GND_OUT	R2.2	GND	22 AWG to component lead	
☐	W17	ESP32.3V3	LCD.VCC	3V3	jumper	
☐	W18	ESP32.3V3	LCD.BL	3V3	jumper	backlight always on
☐	W19	ESP32.GPIO27	SR04.TRIG	TRIG	jumper	
☐	W20	SR04.ECHO	R1.1	ECHO_5V	jumper to component lead	R1 = 2.2 k Ω 1 %
☐	W21	R1.2	R2.1	ECHO_3V	component leads	R2 = 3.3 k Ω 1 %
☐	W22	R1.2	ESP32.GPIO26	ECHO_3V	jumper to component lead	≈ 3.0 V high level
☐	W23	ESP32.GPIO14	SERVO.SIGNAL	SERVO_PWM	jumper	orange/yellow lead
☐	W24	ESP32.GPIO23	LCD.DIN	TFT_MOSI	jumper	
☐	W25	ESP32.GPIO18	LCD.CLK	TFT_SCK	jumper	
☐	W26	ESP32.GPIO33	LCD.CS	TFT_CS	jumper	
☐	W27	ESP32.GPIO17	LCD.DC	TFT_DC	jumper	
☐	W28	ESP32.GPIO16	LCD.RST	TFT_RST	jumper	

Not in the list on purpose: the rear USB-C data lines (power only), the LCD's SD-card pins, and the optional live-link tap (TX0 + GND to a 3.3 V adapter, adapter VCC unconnected — docs/LIVE_LINK.md).

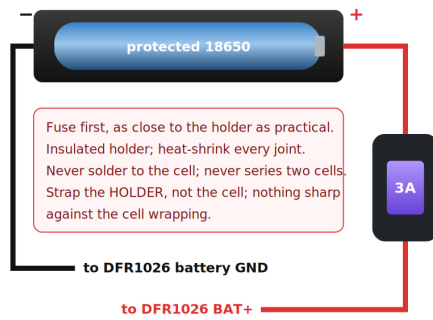
Two details that protect the build

Illustration (not a photo). Resistor colour bands shown for 5-band 1 % parts.

A · ECHO divider (5 V → ≈3 V)



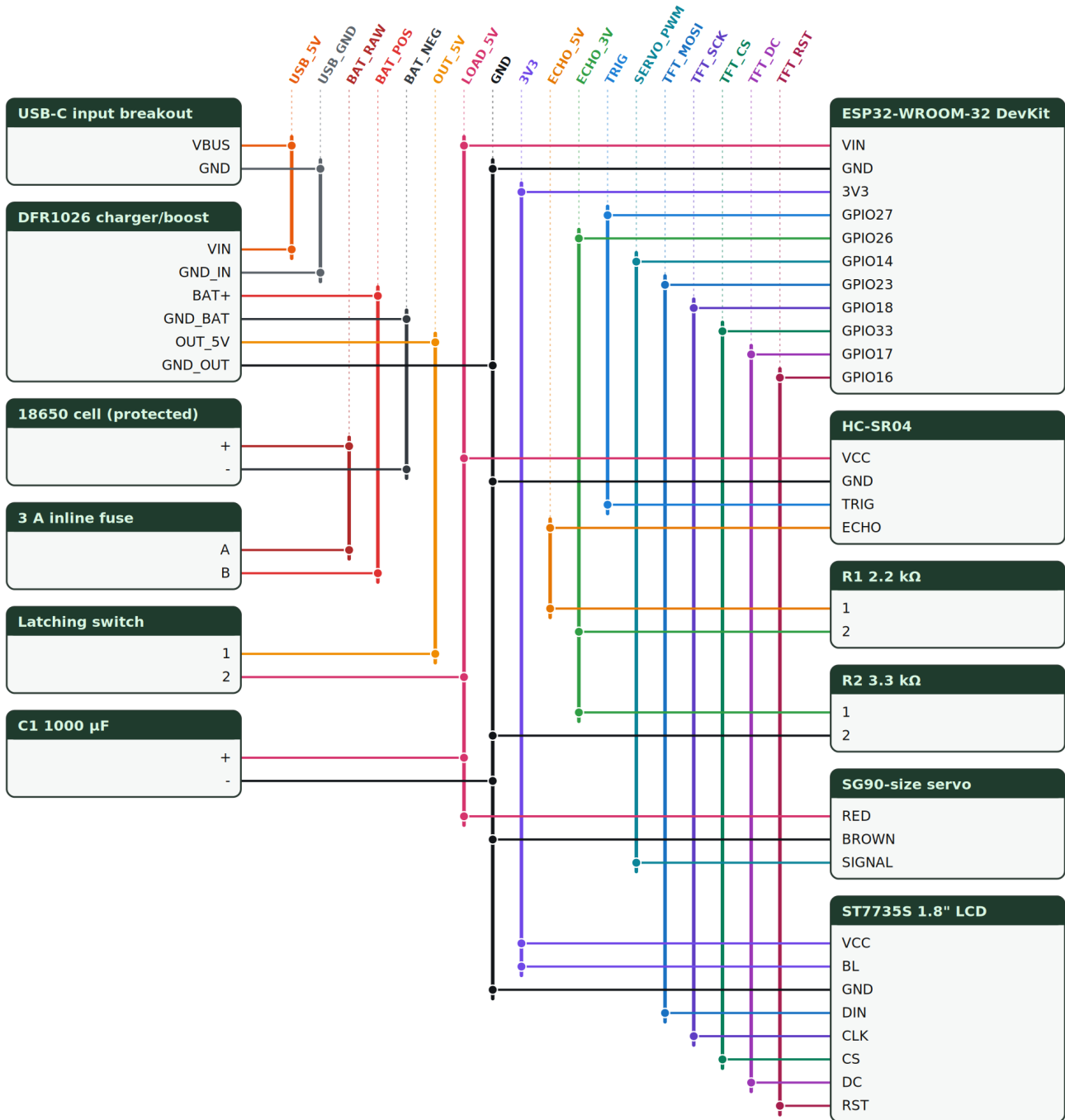
B · 3 A fuse on battery + (mandatory)



Wiring diagram

Radar V5.1 — full wiring (pin → net)

Power and signal. Hover a dot for the net-list row ID.



Generated from the NETLIST table in docs/WIRING.md by scripts/gen_diagrams.py — do not edit by hand.
Dots mark connections; lines that cross without a dot are not connected. Follow labels, not wire colours.

5 · Firmware

- ☐ Switch OFF. Unplug J1 and every jumper from the ESP32 (the whole harness).
- ☐ Connect the ESP32's own USB data port to the computer.
- ☐ EITHER open the web app → **Flash firmware** (Chrome/Edge) and install the release build, OR compile: `arduino-cli compile --upload --profile esp32-core3 -p <port> firmware/Radar_V5`.
- ☐ Unplug the data cable. Re-connect the harness (J1 last), checking labels. Switch ON.
- ☐ Splash shows “EDU SONAR ... NOT a safety device”, then the green fan and the sweep.

6 · Assembly gates

Stage 2 — servo-fit subset

- ☐ Servo through the roof, flange flat, 2× M2×8
- ☐ Horn in hub with 2× M2×6; hub on spline with pointer at 90° (CENTER_ONLY)
- ☐ Keeper fitted with 2× M2×8; hub turns freely 30°-150°
- ☐ Axial play 0.2-0.6 mm (else 0.4 mm shim or change HUB_BOTTOM_ABOVE_ROOF)

Stage 3 — bench power (no ESP32)

- ☐ No shorts with the cell out
- ☐ ≈5 V at OUT_5V and LOAD_5V with switch ON
- ☐ LOAD_5V = 0 V with switch OFF
- ☐ Charges from USB-C with switch OFF
- ☐ OFF→ON restart after 5 s, 60 s, 10 min (open issue)

Stage 4 — flash + bench run

- ☐ Display orientation/colours correct (else TFT_* settings)
- ☐ Sweep direction matches head (else REVERSE_SERVO)
- ☐ Echo at 30 / 100 / 150 cm recorded
- ☐ Travel calibrated, no buzzing at ends

Stage 5 — final assembly

- ☐ LCD in locators, panel on with 4× M2×8
- ☐ ESP32, cell holder, USB-C, DFR1026 secured
- ☐ Sensor header-up, 4× M2×12; mast + M2×20 cross-bolt
- ☐ Sensor cable slack reaches 30° and 150° without tugging
- ☐ Roof on with 4× M2×8; rubber feet; rear label legible

7 · Physical test sheet

Copy results into docs/VALIDATION.md (or open a pull request) — that is how the design becomes validated.

Test	What	Pass criterion	Result	Date / initials
T-PRN-1	Print the fit coupon; tap pilots; try clearances, servo cut-out, switch hole, nut pocket	chosen sizes recorded; CAD updated if needed		
T-PRN-2	Print the servo-fit subset (roof, hub, keeper)	prints without supports or failures		
T-PRN-3	Print P1 (body, front panel) and the sensor head	45° deck and bridges print cleanly		
T-FIT-1	LCD in the front-panel locators; display aperture over the active area	no light leak; active area centred (else set LCD_AA_OFFSET_X/Z)		
T-FIT-2	ESP32 on its supports; cell holder on the anchors; USB-C receptacle flush with the rear opening	plug inserts fully; nothing forced		
T-FIT-3	Hub on spline under the keeper	0.2-0.6 mm axial play; turns freely 30°-150°		
T-FIT-4	Head mast in socket, cross-bolt	no wobble; single orientation		
T-PWR-1	Bench power: no shorts; OUT_5V, LOAD_5V with switch ON/OFF	≈5 V ON; LOAD_5V 0 V OFF		
T-PWR-2	Minimum running current of the complete scanner (servo holding, LCD on)	record mA; compare with the DFR1026 keep-alive (forum reports ~100 mA)		
T-PWR-3	Open issue: OFF → ON after 5 s, 60 s, 10 min; with and without USB-C	output returns without pressing KEY		
T-PWR-4	One-hour run	no drop-outs or resets		
T-PWR-5	Charge with scanner OFF; charge while ON	charging indicated both ways; nothing hot		
T-FW-1	Flash; splash then grid and sweep	as described in ASSEMBLY stage 4		
T-FW-2	Display orientation and colours	correct with INTR_GREENTAB, rotation 1, no inversion — else record the settings that work		
T-FW-3	CENTER_ONLY horn fit; REVERSE_SERVO direction	pointer on 90° tick; screen matches head		
T-FW-4	Measured timing from the serial "# measured" line	record mean step and one-way sweep (estimate: ≈101 ms, ≈4.0 s)		
T-FW-5	Distance at 30 / 100 / 150 cm on the 90° line (flat target)	record reading vs tape measure		
T-CAL-1	Servo calibration against keeper ticks without hitting stops	record SERVO_US_AT_0/180; no buzzing at ends		
T-SWP-1	Powered sweep clearance with the sensor cable fitted	no rubbing, no cable tug at 30° or 150°		
T-RUN-1	Runtime from full charge	record hours		
T-THM-1	Temperatures after 30 min running and during charging (DFR1026, ESP32 regulator, cell)	nothing uncomfortable to touch; record °C if you have a thermometer		
T-LNK-1	Live link: telemetry tap (TX0 + GND only) to a 3.3 V adapter; live console connects at 115 200 baud	readings appear, sweep matches the real head, nothing resets or heats up		
T-LNK-2	Record a session in the console and replay it	replay matches the live view		
T-APP-1	Windows installer + portable build open and connect over the tap	app opens; port picker lists the adapter		
T-APP-2	macOS build opens (after Gatekeeper approval) and connects	as above		
T-FLS-1	Browser flasher (Flash page, Chrome/Edge) installs the release firmware on a bare ESP32 DevKit (harness unplugged)	splash screen appears after re-connecting the harness		
T-REL-1	First tagged release on GitHub carries the source ZIP, manufacturing kit, firmware, desktop apps, SHA256SUMS and attestations	sha256sum -c and gh attestation verify pass		

Sign-off

All gates passed and recorded	☐
Labelled “educational — not a safety device”	☐
Signature / date	